

DEEPAK YADAV Associate Lead Data Scientist

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Experienced Software Engineer and Data Scientist with 5+ years of expertise in developing scalable systems and delivering machine learning solutions across NLP, Computer Vision, Predictive Modeling, and Large Language Models (LLMs). Worked on RAG optimization, knowledge graphs, and LLM-based solutions for text generation, and classification. Skilled in building end-to-end AI pipelines and predictive models, leveraging Python, C++, and cloud platforms. Passionate about solving complex problems through innovative technologies.

SKILL

- **Languages:** Python, SQL, C++, OOPs with Data Structure
- **Big Data & Distributed Systems:** Apache Spark, Akka Streams, Hadoop, Kafka, Redis, RabbitMQ
- **ML Frameworks:** PyTorch, TensorFlow, Hugging Face, ONNX, LangChain, OpenAI.
- **Cloud Platforms & Tools:** Azure ML, Azure Cognitive Search, PostgreSQL, Pinecone, FAISS, Databricks, MIFlow
- **MLOps & DevOps Tools:** Git, Gitlab, DVC, Flask, FastAPI, CI/CD, Docker, Kubernetes
- **ML Techniques:** Decision Trees, Random Forest, Bagging & Boosting, Clustering, Regularization, Anomaly Detection
- **Visualization & Processing:** Matplotlib, Seaborn, Pyspark, Pandas.
- **Domain Expertise:** NRAG, LLM with Fine-tuning, Prompt Engineering, GNN, Transformer

PROFESSIONAL EXPERIENCE

Associate Lead Data Scientist at Bajaj Finserv

05/2023 – present
Pune, India

- Financial Advisory System: Working on a scalable and cost-efficient Retrieval-Augmented Generation (RAG) system to deliver personalized financial product recommendations. Integrated FastAPI, Elasticsearch, and GPT-based models for real-time advice generation. Utilizing Redis for caching, RabbitMQ for asynchronous processing, and Docker/Kubernetes for horizontal scaling. Implemented data preprocessing and EDA to enhance recommendation accuracy.
- Intelligent FAQ and Query Resolution: Developed an intelligent loan document query system that processes PDFs and web content to deliver accurate, context-aware responses. Implemented Retrieval-Augmented Generation (RAG) with vector-based retrieval using GPT API and LLaMA for enhanced document understanding, achieving a 30% improvement in query resolution accuracy across multiple loan products.
- Expense Manager Automation: Built a scalable Expense Manager System for an NBFC using the OneMoney API to fetch and process millions of customer transactions securely. Automated key information extraction (merchant names, payment methods, transaction types) using Azure Databricks, PySpark, and a fine-tuned BERT model, reducing manual effort by 30% and improving extraction accuracy by 22%.

Senior Machine Learning Engineer at SONY

12/2022 – 05/2023

Bangalore, India

- Developed a real-time skeleton-based pose detection system using Spatial-Temporal Graph Convolutional Networks (STGCN) to analyze human motion patterns for surveillance applications. Optimized the model with quantization and pruning for efficient edge device deployment and implemented real-time inference using C++ and TorchScript for serialization.

Data Scientist at Stealth Start-up

05/2022 – 12/2022

Bangalore, India

- Developed a VAE-GAN pipeline to enhance low-quality video frames, detecting drift (foggy/low-light) with a VAE and correcting it using a GAN generator, thereby improving object detection accuracy by 30% via YOLOv5 and optimising inference with TensorRT for real-time performance.

Machine Learning Research Engineer at INRIA, Paris

09/2020 – 05/2022

Paris, France

Project - 1:

- Worked on multilingual datasets and built automated text preprocessing pipelines. Developed LLM-based pipelines to detect radicalisation sentences on social media.

Project - 2:

- Developed a multimodal machine learning pipeline for emotion recognition from video, EEG, ECG, EDA signals, and transcript text data. Achieved high accuracy in classifying human emotions on arousal and valence scales.

Summer Research Intern at National Taiwan University

06/2020 – 08/2020

Hualien, Taiwan

- Worked on the project for Sq2Sq machine translation of indigenous Indian languages (Bengali, Tamil, and Telugu). I've used attention-based LSTM cell-based RNN encoder and decoder architecture to generate Indic language. We obtained an average BLEU score of 4.07, 5.43, and 4.97 for the respective languages trained on 25000 large text corpora.

EDUCATION

Master in Machine Learning and AI at IIIT Bangalore,

Bangalore, India

International Institute of Technology - IIT Bangalore

Bachelor of Engineering(B.E) in Computer Science & Engineering at CU,

Chandigarh, India

Chandigarh University

Major Courses Work - Linear Algebra, Calculus and Vector Space, Discrete Mathematics, Probability & Statistics, Game and Graph Theory, Artificial Intelligence & Machine Learning, Data Structure & Analysis of Algorithms, Object-Oriented Programming,

PUBLICATION

- A. Bahuguna, D. Yadav, A. Senapati, B. N. Saha. **KNN-SVM with Deep Features for COVID-19 Pneumonia Detection from Chest-Xray.** (Accepted for Conference at International Conference on Mathematics and Computing (ICMC) 2022) **[Oral Presentation]** 
- A. Bahuguna, D. Yadav, A. Senapati, B. N. Saha. **A Unified Deep Neuro-Fuzzy Approach for COVID-19 Twitter Sentiment Classification.** (Accepted for Journal at Journal of Intelligent & Fuzzy Systems (JIFS) 2022) **[Oral Presentation]** 

AWARDS

- Selected for the **2020 MOST GASE Global Talent Summer Internship Program(April 2020) (Lab: Artificial Intelligence)** organized by the **Ministry of Science & Technology** and NTU. Selected as one of the 35 students from all over the world.
- Selected for the prestigious **Charpak Exchange (BCS) Scholarship: Awarded by the France Embassy** in India.
- Selected for the **Google Research Week Program(NLU Track)** organized by the Google Research team in India

PROJECTS

Loan Default Predictive Pipeline Using ML

Designed and implemented an end-to-end pipeline integrating feature engineering, SMOTE for class balancing, and RandomForest-based predictive modeling. Achieved 85% accuracy in identifying loan defaults, contributing to improved risk management and decision support for financial institutions.

- **Skills: Feature Engineering, Predictive Modeling, RandomForest, Python, Scikit-learn**

Graph-Based Credit Card Fraud Detection

- Developed a graph-based approach to detect fraudulent transactions in financial data, significantly improving accuracy and recall. Constructed a graph using customer and transaction data, and trained a Graph Neural Network (GNN) to identify patterns indicative of fraud. Deployed the model for real-time fraud detection, ensuring seamless integration with existing financial systems.

Skills: GNN, PyTorch Geometric, NetworkX, FastAPI, Docker.

HIV Classification using Graph Neural Network with Attention

Worked on Graph Convolutional Neural network with attention to classifying HIV using molecule-net data sets. Accuracy: 96.7%, Precision: 38.1% Recall: 43.9%, F1-Score: 40.8. Used GVAE to generate the graph-based molecules

Skill: Pytorch Geometric, Graph Neural Network(GAT), GVAE, Deep Learning.

Fake News Detection Using Mutual Embedding via Graph Neural Network

Worked on social fake news datasets, Extracted news context level features using W2Vec and BERT then merged them with GNN Layers to get social context extraction. Got a TestF1 score of 92%.

Skills: GNN, W2Vec, BERT.

Semantic Segmentation Using U-Net for Satellite Imagery

Developed a U-Net model to segment satellite images into categories such as water, land, roads, buildings, and vegetation. Implemented an end-to-end pipeline with data preprocessing, augmentation, and real-time inference using a Streamlit application. The model is designed for applications like urban planning and environmental monitoring, offering pixel-level accuracy for large-scale land segmentation tasks.

- **Technologies: Python, TensorFlow/Keras, U-Net, OpenCV, Streamlit**

Cartoon Image Generator Using ViT And GPT Based Models

Developed a model combining Vision Transformers (ViT) and Generative Pre-trained Transformers (GPT) to convert images into cartoon-style outputs. Trained on a large dataset of image pairs and deployed as a web application.

Skills: Python, PyTorch, TensorFlow, Transformers (Hugging Face), OpenCV, Flask